Siksha ‘O’ Anusandhan
PROXIMA 2026

Problem Statement - Smart Power Monitoring & Theft Detection System

Theme - Smart Grid Security

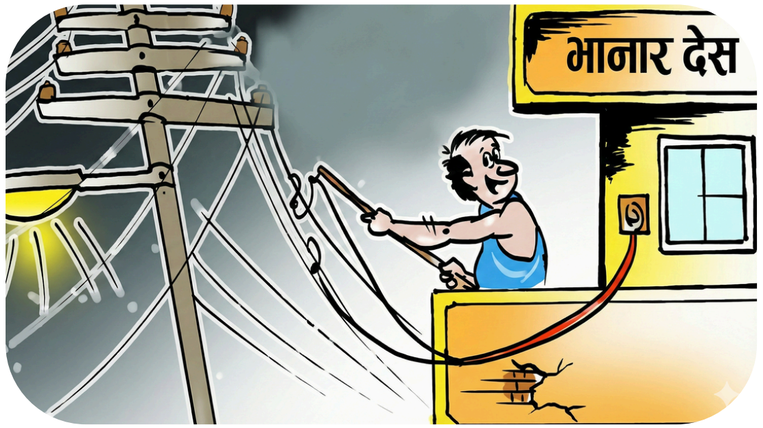
PS Category - Hardware

Team Name - TIFive

PROXIMA 2026

ADDRESSING CRITICAL POWER GRID CHALLENGES





Existing electricity grids face significant issues including pervasive power theft, inaccurate monitoring, and a severe lack of real-time data. These problems result in substantial revenue losses for utilities, unreliable service, and hinder the adoption of smart energy solutions.

Revenue Leakage

Billions lost annually due to unauthorized power consumption and tampering.

Safety Hazards

Dangerous illegal connections increase accident and fire incidents.

Operational Inefficiency

Manual data collection limits real-time insights for load balancing.

Environmental Impact

Energy waste contributes to increased carbon footprint.

Core Sub-Problems

OUR PROPOSED SOLUTION: SMART POWER MONITORING & THEFT DETECTION

IoT-based system designed to revolutionise electricity management and combat power theft effectively.

Real-time Monitoring

Continuous, accurate data on consumption, voltage, and current for instant insights.

Automated Theft Detection

Sophisticated analytics identify anomalies and potential theft attempts instantly.

Enhanced Grid Efficiency

Leverage data for better load balancing, predictive maintenance, and energy optimisation.

TECHNICAL APPROACH

System Architecture Overview

Transformer-Side Node

Monitors supply output at distribution point using isolated voltage and current sensors. Calculates total power delivered and transmits timestamped data to cloud database.

Home-Side Node

Measures actual consumption at individual households. Compares with transformer output to detect anomalies and potential theft through data analytics.

Both nodes interface safely with AC mains using isolation techniques, calculate real power consumption, and stream data over Wi-Fi to Firebase Realtime Database for cloud-based analysis.

Hardware Components & Sensors

- ZMPT101B (Voltage sensor)
- SCT-013 (Current Sensor)
- ESP32 microcontroller



Prototype

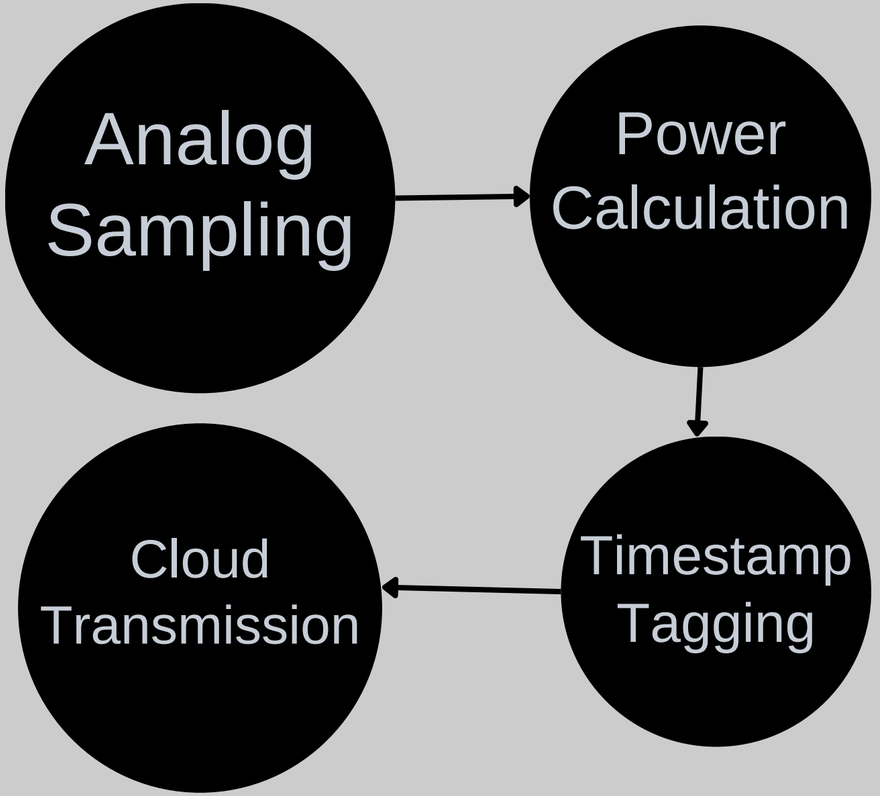
Analog Sampling: ESP32 ADC samples voltage and current at 1-10 kHz rate, capturing AC waveform characteristics

Power Calculation: Real power computed using instantaneous v×i product, RMS values, and power factor correction

Timestamp Tagging: Each measurement tagged with precise timestamp using NTP synchronization for comparison

Cloud Transmission: Data transmitted over Wi-Fi to Firebase Realtime Database with JSON structure

Data Acquisition And Processing



TECHNOLOGY

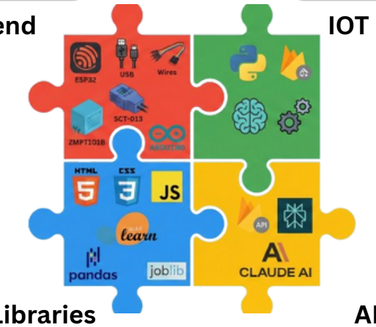
TECHNOLOGY USED

Python, Google Firebase, LLM, ML

AI & Backend

ESP32, USB cables, connections cables, ZMPT101B, SCT-013, Arduino IDE

IOT Integration



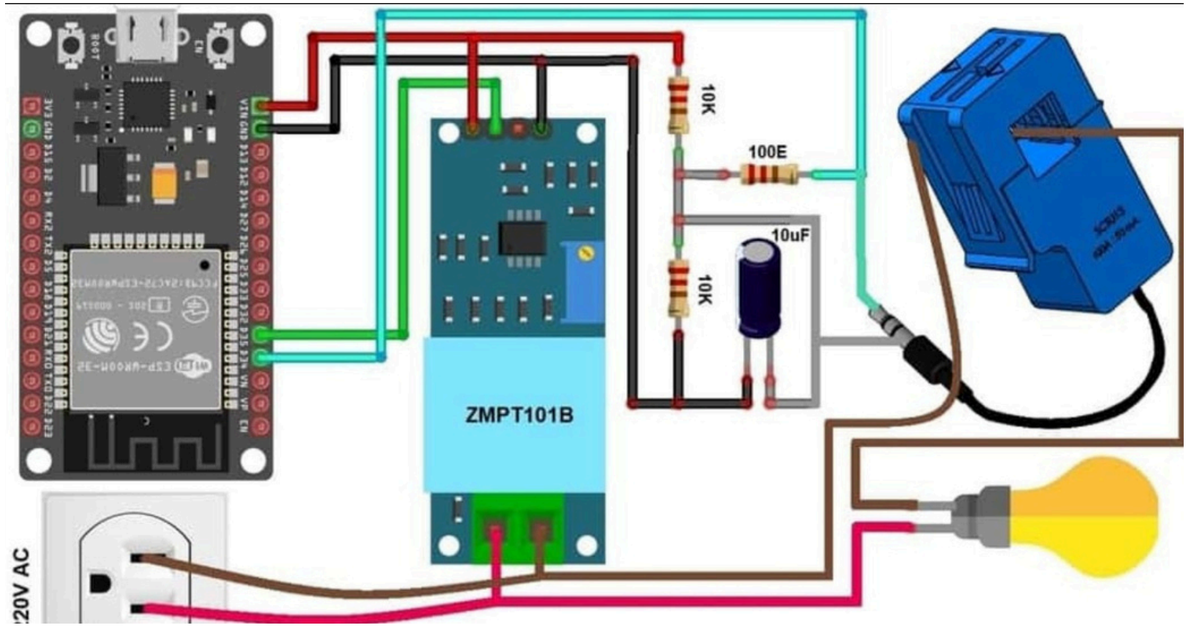
Frameworks & Libraries

HTML, CSS, JS, sci-kit, pandas, joblib

API & Tools

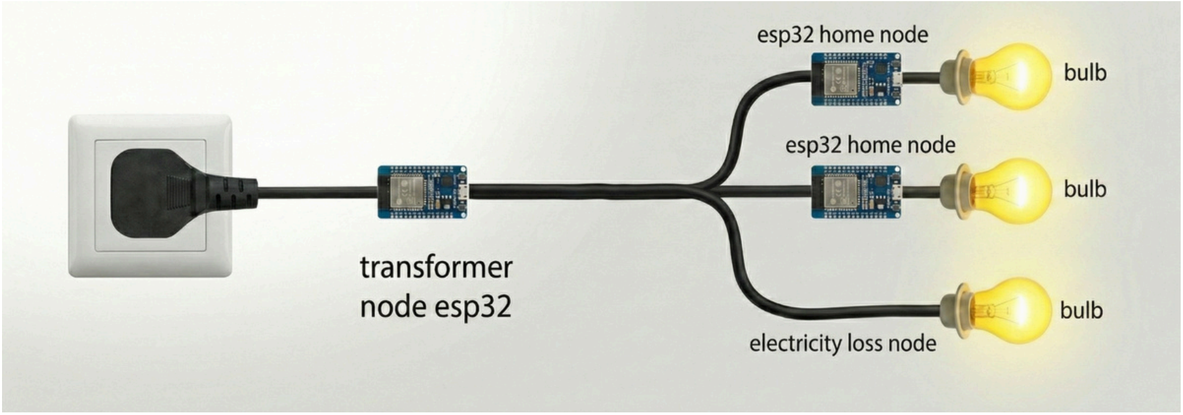
Firebase API, Perplexity AI, Claude

ESP32 CIRCUIT DIAGRAM



DATA FLOW STRUCTURE

SENSORS → ESP32 → CLOUD DATABASE → SERVER PC (for calculation) → CALCULATION UPLOAD TO CLOUD → WEBPAGE

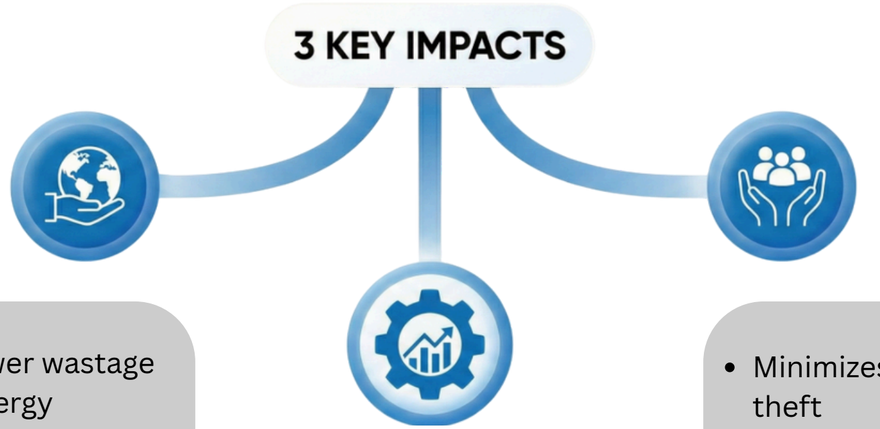


CIRCUIT DIAGRAM

BEFORE & AFTER IMPLEMENTATION: A TRANSFORMATIVE IMPACT

Feature	Before Implementation (Current Scenario)	After Implementation (Your Solution)
Theft Detection	Reactive: Theft is only suspected after a month when the transformer shows huge losses or billing is low.	Proactive: Theft is detected instantly (real-time) when the mismatch occurs.
Inspection Method	Manual Raids: Dangerous, expensive, and often ineffective (thieves hide wires when they see the van).	Data-Driven: Raid teams are sent only to specific locations with digital evidence in hand.
Billing Accuracy	Estimated: Often based on averages if meters are tampered with.	Precise: Based on actual digital logs that are backed up to the cloud.
Transformer Health	Risk of Failure: Overloading goes unnoticed until the transformer fuses blow or it explodes.	Protected: Alerts are sent immediately if total load exceeds the transformer's safety limit.

BENEFITS



- Reduces power wastage
- Improves energy efficiency
- Lowers carbon emissions

- Ensures fair billing
- Improves transparency
- Reliable power supply

- Minimizes electricity theft
- Increases utility revenue
- Low-cost, scalable system

Company Benefits	Customer Benefits
Real-time theft detection	Accurate billing
Reduced energy losses	Stable power supply
ML-based analysis	Transparent usage data
Cost-effective deployment	Faster issue resolution
Smart grid ready	Energy awareness

RESEARCH AND REFERENCES

Sl. No.	Name	Description
1	IoT-Based Smart Energy Meter for Power Theft Detection	This project proposes an IoT-enabled smart energy meter that monitors real-time voltage and current data and transmits it to a cloud platform for detecting abnormal energy consumption patterns indicative of electricity theft. IEEE Reference: A. Jain, S. K. Singh, and R. Kumar, “IoT-based smart energy meter for power theft detection,” International Journal of Engineering Research & Technology (IJERT), vol. 9, no. 6, pp. 123–128, Jun. 2020.
2	Machine Learning–Based Electricity Theft Detection in Smart Grids	This work explores the application of machine learning techniques to identify anomalous power consumption behavior in smart grids, enabling automated detection of electricity theft without manual inspection. IEEE Reference: A. Albert, R. Rajagopal, and R. Sevlian, “Data-driven detection of electricity theft,” IEEE Transactions on Smart Grid, vol. 5, no. 2, pp. 1039–1046, Mar. 2014. doi: 10.1109/TSG.2013.2284354.
3	Anomaly Detection Using Isolation Forest for Energy Systems	This project introduces the Isolation Forest algorithm for detecting anomalies in large datasets and demonstrates its effectiveness in identifying abnormal patterns in energy consumption data. IEEE Reference: F. T. Liu, K. M. Ting, and Z.-H. Zhou, “Isolation Forest,” in Proc. IEEE Int. Conf. Data Mining (ICDM), Pisa, Italy, 2008, pp. 413–422. doi: 10.1109/ICDM.2008.17.
4	ESP32-Based Energy Monitoring System with Cloud Integration	This system utilizes the ESP32 microcontroller along with voltage and current sensors to measure electrical parameters and upload them to a cloud database, enabling remote monitoring and analysis of power usage. **IEEE Reference:** Espressif Systems, “ESP32 Series Datasheet,” Espressif Systems Inc., 2023. [Online]. Available: https://www.espressif.com (Accessed: Jan. 2026).